

The Program in Action

When you run *sinemessage.py*, the output will look like this:

Sine Message, by Al Sweigart al@inventwithpython.com
(Press Ctrl-C to quit.)

What message do you want to display? (Max 39 chars.)

```
> I <3 Programming!
```

I <3 Programming!

I <3 Programming!

I <3 Programming!

I <3 Programming!

I <3 Programming!

I <3 Programming!

I <3 Programming!

I <3 Programming!

I <3 Programming!

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I <3 Programming!

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I <3 Programming!

I <3 Programming!

--snip--

How It Works

The `math.sin()` function in Python's `math` module takes an argument, which we'll call x , and returns another number called the *sine of x* . Several mathematical applications use the sine function; in our program, its purpose is merely to create a neat wave effect. We pass a variable named `step` to `math.sin()`. This variable starts at 0 and increases by 0.25 on each iteration of the main program loop.

We'll use the return value of `math.sin()` to figure out how many spaces of padding we should print on either side of the user's message. Since `math.sin()` returns a floating point number between -1.0 and 1.0, but the minimum amount of padding we want is zero, not a negative value, line 31 adds 1 to the return value of `math.sin()`, making the effective range 0.0 to 2.0. We'll certainly need more than zero to two spaces, so line 31 multiplies this number by a variable named `multiplier` to increase the amount of padding. The product of this multiplication is the number of spaces of padding to add to the left side before printing the user's message.